

Question number	Answer	Mark
7(a)(i)	The only correct answer is D. <i>A is incorrect because pH is a log scale so the difference of 2 is 100 fold</i> <i>B is incorrect because pH is a log scale so the difference of 2 is 100 fold</i> <i>C is incorrect because there are fewer hydrogen ions at pH 8.5 than pH 6.5</i>	(1) COMP

Question number	Answer	Additional guidance	Mark
7(a)(ii)	650 (%)		(1)

Question number	Answer	Additional guidance	Marks
7(a)(iii)	Indicative content: • soil depth increases with distance from the sea (D) • because these are older sand dunes (E) • so more time for humus to build up (E) • organic material increases with distance from the sea (D) • because these are older sand dunes • so more decomposition adds organic material to the sand (E) • due to activity of microorganisms (E) • pH falls with distance from the sea (D) • because more humus is added to the sand (E) • more rain (over the years) washes out minerals (E) • percentage of bare rock decreases with distance from sea (D) • because pioneer species break it down (E) • because as more soil is produced, it covers the bare rock (E) • number of different species of plant increase with distance from sea (D) • because the soil {is becoming more fertile / salinity is decreasing} (E) • organic material holds more moisture than sand alone (E) • types of plant change with distance from sea (D) • as soil becomes more favourable for more plant species (E) • more plants improve soil conditions further (E)	Level 1 : 1 mark = 1 description of {data / succession} 2 marks = 3 descriptions {of data / succession} OR 1 description of data + an explanation Level 2 : 3 marks = 2 descriptions of the data and an explanation of both 4 marks = 3 descriptions of the data and all three explained Level 3 : 5 marks = 4 descriptions of the data and all 4 explained 6 marks = 5 descriptions of the data and all five explained	(6)

Question number	Answer	Additional guidance	Mark
7(b)(i)	A description that includes the following points: • NPP above ground is always greater than the NPP below the ground (1) • the NPP above and below the ground depends on the type of sand dune (1) • fixed sand dunes have the highest (total) NPP / shifting sand dunes have the lowest (total) NPP (1) • shifting sand dune has the largest {percentage difference / ratio} between NPP above the ground and NPP below the ground (1)		(3)

Question number	Answer	Additional guidance	Mark
7(b)(ii)	An explanation that includes four of the following points: • because {light-independent reactions / Calvin cycle} produce {GALP / glucose / hexose} (1) • {GALP / glucose / hexose} used to produce sucrose (1) • {GALP / glucose / hexose} used to produce amino acids (1) • (sucrose / amino acids) transported in the phloem to the {roots / rhizomes} (1) • {glucose / sucrose / amino acids} used to synthesise {organic material / biomass / named organic molecule / NPP} (1)	DO NOT ACCEPT glucose transported	(4)